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A YANG Data Model for Resource Public Key Infrastructure (RPKI) to Router Protocol

draft-ietf-sidrops-rtr-yang-13

Abstract

This document defines a YANG data model~~s~~ for managing Resource Public Key Infrastructure (RPKI) to Router Protocol (RFC 6810 and RFC 8210).

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Commenté [MB1]: Please reflect the udpated title in the modules.

Commenté [MB2]: Do we need this in the abstract?

The data model also cover 8210bis.

I suggest to delete here to avoid inconsistency with the statement in the introduction.

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1. Introduction

[RFC6810] and [RFC8210] describe a protocol to deliver Resource Public Key Infrastructure (RPKI) prefix origin data and router keys from a trusted cache server to a router, referred to as the RPKI to Router (RTR) protocol.

[I-D.ietf-sidrops-8210bis] describes version 2 of the RTR protocol, which adds a new Autonomous System Provider Authorization (ASPA) PDU type.

This document defines a YANG [RFC7950] data model~~s~~ for managing RTR protocol ([RFC6810], [RFC8210], and [I-D.ietf-sidrops-8210bis]).

Commenté [MB3]: Different from the statement in the abstract.

2. Terminology and Notation Conventions

2.1. Requirements Notation

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

2.2. Terminology

The following terms are defined in [RFC7950] and are not redefined here:

* augment
* data model
* data node
* module
* namespace
* ~~YANG~~

2.3. Tree Diagrams

Tree diagrams used in this document follow the notation defined in [RFC8340].

3. Data Model Overview

This document defines two YANG modules: "ietf-rpki-rtr" (Section xxx) and "ietf-rpki-table" (Section XXX).

The "ietf-rpki-rtr" YANG module provides the methods for managing RTR protocol. It includes:

* Connectivity parameters, such as RPKI cache server IP address and destination port number.

* Session parameters, such as refresh interval, retry interval, expire interval, and hold interval.

* Session status and statistics, such as session ID, serial number, and number of received and transmitted messages.

The "ietf-rpki-table" YANG module provides the methods for managing records of RTR protocol and the corresponding state hash which is a hash value used in the Canonical Cache Representation (CCR) content [I-D.ietf-sidrops-rpki-ccr]. It includes:

* Validated ROA Payload (VRP) records.

* Router Key records.

* ASPA records.

* CCR state hash, which is optional and is used to verify the integrity and consistency of RPKI data originating from the RPKI cache.

Commenté [MB4]: Help reader identify where the module is defined.

Commenté [MB5]: I would define a feature for aspa support.

Commenté [MB6]: -- Shouldn't this be defined as a feature?

4. RPKI to Router YANG Module

4.1. Tree View

The full tree of the "ietf-rpki-rtr" YANG module is provided in Appendix A. The following subsections list the subtree structures.

4.1.1. Overall Structure

The overall tree structure of the "ietf-rpki-rtr" YANG module is shown in Figure 1.

The "ietf-rpki-rtr" YANG module augments the core routing YANG module "ietf-routing" specified in [RFC8349] for managing RTR protocol sessions. Specifically, this YANG module augments "/rt:routing/ rt:control-plane-protocols/rt:control-plane-protocol".

```
module: ietf-rpki-rtr
  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol:
    +--rw rpki-rtr
       +--rw sessions
          +--rw session* [server-address]
             +--rw server-address           inet:ip-address
             +--rw server-port?             inet:port-number
             +--rw local-address?           union
             +--rw local-port?              inet:port-number
             +--rw enabled?                 boolean
             +--rw preference?              uint32
             +--rw description?             string
             +--rw reconnect-interval?      uint32
             +--ro session-state?           enumeration
             +--rw enable-authentication?   boolean
             +--rw authentication
             |  ...
             +--rw vrp-limit
             |  ...
             +--rw aspa-limit
             |  ...
             +--ro statistics
             |  ...
             +--ro connection-data
             |  ...
             +--rw protocol-data
             |  ...
             +--ro pdu-counters
             |  ...
             +--ro error-pdu-counters
                ...
```

Figure 1: Overall RPKI to Router Tree Structure

The "ietf-rpki-rtr" YANG module has a root container 'rpki-rtr', which has a subcontainer 'sessions'. The container 'sessions' contains a list 'session', where each entry represents a session configuration and state to a specific RPKI cache server.

4.1.2. RTR Session ~~Configuration~~Structure

The RTR session configuration structure for the list 'session' is shown in Figure 2.

Commenté [MB7]: As this can be used for state retrieval as well.

```
module: ietf-rpki-rtr
  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol:
    +--rw rpki-rtr
       +--rw sessions
          +--rw session* [server-address]
             +--rw server-address           inet:ip-address
             +--rw server-port?             inet:port-number
             +--rw local-address?           union
             +--rw local-port?              inet:port-number
             +--rw enabled?                 boolean
             +--rw preference?              uint32
             +--rw description?             string
             +--rw reconnect-interval?      uint32
             +--rw enable-authentication?   boolean
             +--rw authentication
             |  ...
             +--rw vrp-limit
             |  ...
             +--rw aspa-limit
             |  ...
             +--rw protocol-data
             |  ...
```

Figure 2: RTR Session Configuration Subtree Structure

The ~~configuration~~ data nodes for the list 'session' are described as follows:

'server-address': Indicates the IP address of the RPKI cache server.

'server-port': Indicates the port number the server listens on.

'local-address': Indicates the source IP address used when initiating a RTR session connection locally.

'local-port': Indicates the source port number used when initiating a RTR session connection locally.

'enabled': Indicates the management switch for the RTR session.

'preference': Indicates the priority for connecting to the RPKI cache server.

'description': Indicates the description information for the RTR session.

Commenté [MB8]: I guess local/server addresses should belong to the same address family. I would say so in an Operational Considerations Section and indicate that any mismatch is considered as an anomaly.

'reconnect-interval': Indicates the reconnection interval after session disconnection.

'enable-authentication': Indicates the switch to enable connection authentication for the RTR session.

'authentication': Indicates the ~~configuration~~ information of connection authentication for the RTR session, as shown in Figure 3.

```
module: ietf-rpki-rtr
  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol:
    +--rw rpki-rtr
       +--rw sessions
          +--rw session* [server-address]
             ...
             +--rw authentication
             |  +--rw (option)?
             |     +--:(md5)
             |     |  +--rw md5-password?   ianach:crypt-hash
             |     +--:(ssh)
             |     |  +--rw client-identity
             |     |  |  +--rw username?  string
             |     |  |  +--rw public-key!
             |     |  |                   {client-ident-publickey}?
             |     |  |  +--rw password!
             |     |  |                   {client-ident-password}?
             |     |  |  +--rw hostbased!
             |     |  |                   {client-ident-hostbased}?
             |     |  |  +--rw none?      empty {client-ident-none}?
             |     |  |  +--rw certificate!
             |     |  |                   {sshcmn:ssh-x509-certs}?
             |     |  +--rw server-authentication
             |     |  |  +--rw ssh-host-keys!
             |     |  |  +--rw ca-certs!  {sshcmn:ssh-x509-certs}?
             |     |  |  +--rw ee-certs!  {sshcmn:ssh-x509-certs}?
             |     |  +--rw transport-params
             |     |  |                {sshcmn:transport-params}?
             |     |  +--rw keepalives!   {ssh-client-keepalives}?
             |     |     +--rw max-wait?       uint16
             |     |     +--rw max-attempts?   uint8
             |     +--:(tcp-ao-keychain)
             |        +--rw keychain-name?  key-chain:key-chain-ref
             +--rw vrp-limit
             |  ...
             +--rw aspa-limit
             |  ...
```

Commenté [MB9]: Please indicate the unit.

Commenté [MB10]: This can be used for retrieval as well.

Figure 3: RTR Session Authentication Subtree Structure

'vrp-limit' (Figure 4): Indicates the limit ~~configuration~~ on the number of VRP ~~record~~ records received from the RTR session.

'aspa-limit' (Figure 4): Indicates the limit ~~configuration~~ on the number of ASPA ~~record~~ records received from the RTR session.

```
module: ietf-rpki-rtr
  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol:
     +--rw rpki-rtr
        +--rw sessions
           +--rw session* [server-address]
              ...
              +--rw authentication
              |  ...
              +--rw vrp-limit
              |  +--rw max-number?              uint64
              |  +--rw threshold-percentage?    rt-types:percentage
              |  +--rw over-threshold-action?   enumeration
              +--rw aspa-limit
              |  +--rw max-number?              uint64
              |  +--rw threshold-percentage?    rt-types:percentage
              |  +--rw over-threshold-action?   enumeration
```

Figure 4: RTR Session Received Record Limit Subtree Structure

As depicted ~~In~~ in Figure 4, 'vrp-limit' and 'aspa-limit' have the same structure, both having ~~the~~ a maximum number of records allowed to be received from the RTR session ('max-number'), ~~the~~ threshold percentage for record maximum number ('threshold-percentage'), and handling strategy when exceeding the record threshold ('over-threshold-action').

'protocol-data': Indicates the protocol parameters of the RTR session. The ~~configurable parameters~~ related data nodes include the preferred protocol version ('preferred-protocol-version') and the intervals for refresh ('refresh-interval'), retry ('retry-interval'), expire ('expire-interval'), and hold ('hold-interval') timers, as shown in Figure 5.

Commenté [MB11]: Indicate the units

```
module: ietf-rpki-rtr
  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol:
    +--rw rpki-rtr
       +--rw sessions
          +--rw session* [server-address]
             ...
             +--rw protocol-data
             |  +--rw preferred-protocol-version?   uint8
             |  +--rw refresh-interval?             uint32
             |  +--rw retry-interval?               uint32
             |  +--rw expire-interval?              uint32
             |  +--rw hold-interval?                uint32
             |  ...
             ...
```

Figure 5: RTR Session Protocol Data Configuration Subtree Structure

4.1.3. RTR Session State

The RTR session state structure for the list 'session' is shown in Figure 6.

```
module: ietf-rpki-rtr
  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol:
    +--rw rpki-rtr
       +--rw sessions
          +--rw session* [server-address]
             +--ro session-state?           enumeration
             +--ro statistics
             |  ...
             +--ro connection-data
             |  ...
             +--rw protocol-data
             |  ...
             +--ro pdu-counters
             |  ...
             +--ro error-pdu-counters
                ...
```

Figure 6: RTR Session State Subtree Structure

The state data nodes for the list 'session' are described as follows:

'session-state': Indicates the connection status of the RTR session.

'statistics': Indicates the statistics of RPKI records or messages received from the server of the RTR session, as shown in Figure 7.

```
module: ietf-rpki-rtr
  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol:
    +--rw rpki-rtr
       +--rw sessions
          +--rw session* [server-address]
             ...
             +--ro statistics
             |  +--ro total-vrp-records?  yang:zero-based-counter64
             |  +--ro ipv4-vrp-records?   yang:zero-based-counter64
             |  +--ro ipv6-vrp-records?   yang:zero-based-counter64
             |  +--ro router-key-records? yang:zero-based-counter64
             |  +--ro aspa-records?       yang:zero-based-counter64
             |  +--ro in-total-messages?  yang:zero-based-counter64
             |  +--ro out-total-messages? yang:zero-based-counter64
             +--ro connection-data
             |  ...
             +--rw protocol-data
             |  ...
             +--ro pdu-counters
             |  ...
             +--ro error-pdu-counters
                ...
```

Figure 7: RTR Session Statistics Subtree Structure

'connection-data': Indicates the state information relating to the session connection with the RPKI cache server, as shown in Figure 8.

Commenté [MB12]: Any reason why the structure does not track discontinuity time since when the counters suffered a discontinuity?

```
module: ietf-rpki-rtr
  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol:
    +--rw rpki-rtr
       +--rw sessions
          +--rw session* [server-address]
             ...
             +--ro statistics
             |  ...
             +--ro connection-data
             |  +--ro flaps?                        uint32
             |  +--ro last-session-up-down?         yang:date-and-time
             |  +--ro last-update-sync?             yang:date-and-time
             |  +--ro last-full-sync?               yang:date-and-time
             |  +--ro last-serial-query?            yang:date-and-time
             |  +--ro last-reset-query?             yang:date-and-time
             |  +--ro last-eod-received?            yang:date-and-time
             |  +--ro last-config-change?           yang:date-and-time
             |  +--ro last-error?                   yang:date-and-time
             |  +--ro last-connection-error?        yang:date-and-time
             |  +--ro last-connection?              yang:date-and-time
             |  +--ro error-reason?                 string
             +--rw protocol-data
             |  ...
             +--ro pdu-counters
             |  ...
             +--ro error-pdu-counters
                ...
```

Figure 8: RTR Session Connection Data Subtree Structure

'protocol-data': Indicates the state parameters of the RTR protocol of the session. The state parameters include the negotiated protocol version ('protocol-version'), the scheduled time points for refresh ('refresh-time'), response ('response-time'), record lifetime ('record-lifetime'), and hold ('hold-time'), as well as the session identifier ('session-id') and serial numbers ('serial-full' and 'serial-incremental'), as shown in Figure 9. The configurable protocol parameters in 'protocol-data' are described in Section 4.1.2.

```
module: ietf-rpki-rtr
  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol:
    +--rw rpki-rtr
       +--rw sessions
          +--rw session* [server-address]
             ...
             +--ro statistics
             |  ...
             +--ro connection-data
             |  ...
             +--rw protocol-data
             |  ...
             |  +--ro protocol-version?   uint8
             |  +--ro refresh-time?       yang:date-and-time
             |  +--ro response-time?      yang:date-and-time
             |  +--ro record-lifetime?    yang:date-and-time
             |  +--ro hold-time?          yang:date-and-time
             |  +--ro session-id?         uint16
             |  +--ro serial-full?        uint32
             |  +--ro serial-incremental? uint32
             +--ro pdu-counters
             |  ...
             +--ro error-pdu-counters
                ...
```

Figure 9: RTR Session Protocol Data State Subtree Structure

'pdu-counters': Indicates the send and receive statistics of various types of RTR PDUs, as shown in Figure 10.

```
module: ietf-rpki-rtr
  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol:
    +--rw rpki-rtr
       +--rw sessions
          +--rw session* [server-address]
             ...
             +--ro statistics
             |  ...
             +--ro connection-data
             |  ...
             +--rw protocol-data
             |  ...
             +--ro pdu-counters
             |  +--ro serial-notify?    yang:zero-based-counter64
             |  +--ro serial-query?     yang:zero-based-counter64
             |  +--ro reset-query?      yang:zero-based-counter64
             |  +--ro cache-response?   yang:zero-based-counter64
             |  +--ro ipv4-prefix?      yang:zero-based-counter64
             |  +--ro ipv6-prefix?      yang:zero-based-counter64
             |  +--ro end-of-data?      yang:zero-based-counter64
             |  +--ro cache-reset?      yang:zero-based-counter64
             |  +--ro router-key?       yang:zero-based-counter64
             |  +--ro aspa?             yang:zero-based-counter64
             +--ro error-pdu-counters
                ...
```

Figure 10: RTR Session PDU Counters Subtree Structure

'error-pdu-counters': Indicates counters of Error Report PDUs exchanged with the RPKI cache server, by error code, as shown in Figure 11.

Commenté [MB13]: -- Why isn't this modeled as a list indexed per pdu-type? I would define an IANA-maintained model that mirrors https://www.iana.org/assignments/rpki#rpki-rtr-pdu and use that type here.

This would future proof as future PDU can inherit the same structure.

```
module: ietf-rpki-rtr
  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol:
    +--rw rpki-rtr
       +--rw sessions
          +--rw session* [server-address]
             ...
             +--ro statistics
             |  ...
             +--ro connection-data
             |  ...
             +--rw protocol-data
             |  ...
             +--ro pdu-counters
             |  ...
             +--ro error-pdu-counters
                +--ro corrupt-data?      yang:zero-based-counter64
                +--ro internal-error?    yang:zero-based-counter64
                +--ro no-data-available? yang:zero-based-counter64
                +--ro invalid-request?   yang:zero-based-counter64
                +--ro unsupported-protocol-version?
                |                        yang:zero-based-counter64
                +--ro unsupported-pdu-type?
                |                        yang:zero-based-counter64
                +--ro withdrawal-unknown-record?
                |                        yang:zero-based-counter64
                +--ro duplicate-announcement-received?
                |                        yang:zero-based-counter64
                +--ro unexpected-protocol-version?
                |                        yang:zero-based-counter64
                +--ro aspa-provider-list-error?
                |                        yang:zero-based-counter64
                +--ro transport-failure?  yang:zero-based-counter64
                +--ro ordering-error?      yang:zero-based-counter64
                +--ro cache-restart?       yang:zero-based-counter64
                +--ro cache-shutdown?      yang:zero-based-counter64
```

Figure 11: RTR Session Error PDU Counters Subtree Structure

Commenté [MB14]: Similar to the previous comment, I would simplify the structure here and define a type based on an IANA-maintained module that mirrors https://www.iana.org/assignments/rpki#rpki-rtr-error

The structure would a list of error/counter with error as an index.

4.2. YANG Module

This YANG module has normative references to [RFC6810], [RFC7317], [RFC8177], [RFC8210], [RFC8294], [RFC8343], [RFC8349], [RFC9644], [RFC9911], and [I-D.ietf-sidrops-8210bis].

```
  Note to the RFC Editor:— _ Note that the "RFC ZZZZ" in this YANG module
must be replaced by
     the publication version of [I-D.ietf-sidrops-8210bis] and this
     note is to be removed after the publication of
     [I-D.ietf-sidrops-8210bis].

   <CODE BEGINS> file "ietf-rpki-rtr@2026-10-15.yang"
   module ietf-rpki-rtr {
     yang-version 1.1;
     namespace "urn:ietf:params:xml:ns:yang:ietf-rpki-rtr";
     prefix rpki-rtr;

     import ietf-yang-types {
       prefix yang;
       reference
         "RFC 9911: Common YANG Data Types, Section 3";
     }
     import ietf-inet-types {
       prefix inet;
       reference
         "RFC 9911: Common YANG Data Types, Section 4";
     }
     import ietf-routing {
       prefix rt;
       reference
         "RFC 8349: A YANG Data Model for Routing Management
                    (NMDA Version)";
     }
     import ietf-routing-types {
       prefix rt-types;
       reference
         "RFC 8294: Common YANG Data Types for the Routing Area";
     }
     import iana-crypt-hash {
       prefix ianach;
       reference
         "RFC 7317: A YANG Data Model for System Management";
     }
     import ietf-ssh-client {
       prefix sshc;
       reference
         "RFC 9644: YANG Groupings for SSH Clients and SSH Servers";
     }
     import ietf-interfaces {
       prefix if;
       reference
         "RFC 8343: A YANG Data Model for Interface Management";
     }
     import ietf-key-chain {
```

Commenté [MB15]: Also, consider having a note for RFC XXXX

Commenté [MB16]: Add section where the module is defined.

Commenté [MB17]: Idem

Commenté [MB18]: Please update to use the IANA registry as authoritative source. That RFC only define the initial version.

Refer to RFC9907 for more guidance on the matter.

```
      prefix key-chain;
      reference
        "RFC 8177: YANG Data Model for Key Chains";
    }

    organization
      "IETF Secure Inter-Domain Routing Operations (SIDROPS)
       Working Group";
    contact
      "WG Web:   <https://datatracker.ietf.org/wg/sidrops/>
       WG List:  SIDROPS <mailto:sidrops@ietf.org>

       EditorAuthor:   Yisong Liu
                 <mailto:liuyisong@chinamobile.com>
       AuthorEditor:   Changwang Lin
                 <mailto:linchangwang.04414@h3c.com>
       AuthorEditor:   Haibo Wang
                 <mailto:rainsword.wang@huawei.com>
       AuthorEditor:   Jishnu Roy
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       AuthorEditor:   Jeffrey Haas
                 <mailto:jeffrey.haas@hpe.com>
       AuthorEditor:   Hongwei Liu
                 <mailto:liu.hongwei3@zte.com.cn>
       AuthorEditor:   Di Ma
                 <mailto:madi@zdns.cn>";
    description
      "This module describes a YANG model module for the Resource Public
       Key Infrastructure (RPKI) to Router (RTR) protocol management.

       The key words 'MUST', 'MUST NOT', 'REQUIRED', 'SHALL', 'SHALL
       NOT', 'SHOULD', 'SHOULD NOT', 'RECOMMENDED', 'NOT RECOMMENDED',
       'MAY', and 'OPTIONAL' in this document are to be interpreted as
       described in BCP 14 (RFC 2119) (RFC 8174) when, and only when,
       they appear in all capitals, as shown here.

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       All revisions of IETF and IANA published modules can be found
       at the YANG Parameters registry group
```

Commenté [MB19]: Follow 9907 template

Commenté [MB20]: Please refer to RFC9907 for the use of model/module

a mis en forme : Anglais (États-Unis)

```
        (https://www.iana.org/assignments/yang-parameters).

        This version of this YANG module is part of RFC XXXX; see the
        RFC itself for full legal notices.";

      revision 2026-10-15 {
        description
          "Initial Version";
        reference
          "RFC XXXX: A YANG Data Model for RPKI to Router Protocol";
      }

      identity rpki-rtr {
        base rt:routing-protocol;
        description
          "RTR protocol.";
      }

      grouping records-limit {
        description
          "Limit of records that can be received from the an RPKI
           cache server.";
        leaf max-number {
          type uint64;
          description
            "Configures Indicates the maximum number of records that can
be
             received from the RPKI cache server.";
        }
        leaf threshold-percentage {
          type rt-types:percentage;
          units "percent";
          description
            "Configures Specifies the threshold percentage for record
maximum
             number.";
        }
        leaf over-threshold-action {
          type enumeration {
            enum alert-only {
              description
                "Generates alert messages.";
            }
            enum discard {
              description
                "Discards excess records.";
            }
            enum reconnect {
              description
                "Disconnects with the RPKI cache server,
```

Commenté [MB21]: As there might be many

Commenté [MB22]: As this can be used for retrieval as well.

Commenté [MB23]: Consider expanding the description to indicate how the threshold is used.

Commenté [MB24]: Are we confident these are the definite list of possible actions?

If not, please transform this into identities.

Commenté [MB25]: When this mode is used, how the alert will be generated? How can that can be controlled/managed?

```
                and tries to reconnect after reconnection
                timer expires.";
            }
            enum idle-forever {
              description
                "Disconnects with the RPKI cache server
                 forever.";
            }
          }
          description
            "The action to be taken when record number exceeds
             threshold.";
        }
      }

      augment "/rt:routing/rt:control-plane-protocols/"
            + "rt:control-plane-protocol" {
        when "derived-from-or-self(rt:type, 'rpki-rtr')" {
          description
            "This augmentation is valid for a routing protocol
             instance of RTR.";
        }
        description
          "RTR protocol augmentation of ietf-routing module
           control-plane-protocol.";
        container rpki-rtr {
          description
            "~~Configuration~~ Data nodes ~~parameters~~ for the RTR protocol.";
          container sessions {
            description
              "Parameters of RPKI sessions to cache servers.";
            list session {
              key "server-address";
              description
                "Each entry contains parameters for an~~a~~ RPKI session
                 identified by the 'server-address' key.";
              leaf server-address {
                type inet:ip-address;
                mandatory true;
                description
                  "The IP address of the RPKI cache server resembling
                   a session.";
              }
              leaf server-port {
                type inet:port-number;
                description
                  "The remote port number for the connection
                   to the RPKI cache server.";
```

Commenté [MB26]: Add the prefix

Commenté [MB27]: What is the default value if not present?

```
            }
            leaf local-address {
              type union {
                type inet:ip-address;
                type if:interface-ref;
              }
              description
                "The local IP (either IPv4 or IPv6) address to use for
                 the connection to the RPKI cache server. This may be
                 expressed as either an IP address or reference to the
                 name of an interface.";
            }
            leaf local-port {
              type inet:port-number;
              description
                "The local port number for the connection
                 to the RPKI cache server.";
            }
            leaf enabled {
              type boolean;
              default "true";
              description
                "Whether the RPKI cache server is enabled.";
            }
            leaf preference {
              type uint32;
              description
                "The router's preference to connect to that cache.
                 The lower the value, the more preferred.";
            }
            leaf description {
              type string;
              description
                "Records a Textual textual description of the RPKI cache
server.";
            }
            leaf reconnect-interval {
              type uint32 {
                range "1..30000";
              }
              units "minutes";
              description
                "Time interval for the reconnection timer.";
            }
            leaf session-state {
              type enumeration {
                enum idle {
                  description
                    "The session is down.";
```

Commenté [MB28]: Consider adding some text to indicate that this should the same address family. Any mismatch is to be considered as anomaly.

a mis en forme : Anglais (États-Unis)

Commenté [MB29]: What is the default value?

Commenté [MB30]: Add a reference statement to rfc8210/#section-10

Commenté [MB31]: Can we please add a reference (RFC/section) where that range is defined?

Commenté [MB32]: Add a reference where the state are defined.

```
                }
                enum connect {
                  description
                    "The session is waiting for the underlying
                     transport session to be established.";
                }
                enum establish {
                  description
                    "The session is up.";
                }
                enum ex-incr {
                  description
                    "Incremental update of records in progress.";
                }
                enum ex-full {
                  description
                    "Full update of records in progress.";
                }
              }
              config false;
              description
                "The session state.";
            }
            leaf enable-authentication {
              type boolean;
              default "false";
              description
                "Specifies ~~Whether~~ whether the session is secured. It is
enabled when set to true, and disable when set to false.";
            }
            container authentication {
              when "../enable-authentication = 'true'";
              description
                "Container for describing how a particular session
                 is to be secured.";
              choice option {
                description
                  "Choice for session securing methods.";
                case md5 {
                  leaf md5-password {
                    type ianach:crypt-hash;
                    description
                      "The password for ~~md5~~ MD5 authentication.";
                  }
                  description
                    "Uses TCP-MD5 to secure the session.";
                }
                case ssh {
                  uses sshc:ssh-client-grouping {
```

Commenté [MB33]: Please use meaningful names

Commenté [MB34]: I guess this is for exchange? If so, please use the full name. Refer to 9907 for the guidance.

Commenté [MB35]: Add some text to explain why this is disabled by default. A reference would suffice.

Commenté [MB36]: Maybe add the following to the Security Considerations:

NEW:
The module supports MD5 to accommodate the installed RTR base. MD5 suffers from the security weaknesses discussed in Section 2 of [RFC6151] and Section 2.1 of [RFC6952].

```
                    reference
                      "RFC 9644: YANG Groupings for SSH Clients and
                                 SSH Servers";
                  }
                  description
                    "Uses SSH to secure the session.";
                }
                case tcp-ao-keychain {
                  leaf keychain-name {
                    type key-chain:key-chain-ref;
                    description
                      "Name of key chain.";
                    reference
                      "RFC 8177: YANG Data Model for Key Chains";
                  }
                  description
                    "Uses key-chain to secure the session.";
                }
              }
            }
            container vrp-limit {
              description
                "Limit of Validated ROA Payload (VRP) records that can
be
                 received from the RPKI cache server.";
              uses records-limit;
            }
            container aspa-limit {
              description
                "Limit of Autonomous System Provider Authorization
                 (ASPA) records that can be received from the RPKI
                 cache server.";
              uses records-limit;
            }
            container statistics {
              config false;
              description
                "Statistics of the RPKI cache server.";
              leaf total-vrp-records {
                type yang:zero-based-counter64;
                description
                  "The total number of Validated ROA Payloads
                   received from the RPKI cache server.";
              }
              leaf ipv4-vrp-records {
                type yang:zero-based-counter64;
                description
                  "The number of Validated ROA Payloads for IPv4
                   prefixes received from the RPKI cache server.";
```

Commenté [MB37]: To ensure modularity, maybe add if-feature aspa

```
        }
        leaf ipv6-vrp-records {
          type yang:zero-based-counter64;
          description
            "The number of Validated ROA Payloads for IPv6
             prefixes received from the RPKI cache server.";
        }
        leaf router-key-records {
          type yang:zero-based-counter64;
          description
            "The number of Router Keys received from the RPKI
             cache server.";
        }
        leaf aspa-records {
          type yang:zero-based-counter64;
          description
            "The number of ASPAs received from the RPKI
             cache server.";
        }
        leaf in-total-messages {
          type yang:zero-based-counter64;
          description
            "The total number of messages received from the
             RPKI cache server.";
        }
        leaf out-total-messages {
          type yang:zero-based-counter64;
          description
            "The total number of messages transmitted to the
             RPKI cache server.";
        }
      }
      container connection-data {
        config false;
        description
          "State information relating to the connection
           with the RPKI cache server.";
        leaf flaps {
          type uint32;
          description
            "Count for number of flaps observed on the
             session.";
        }
        leaf last-session-up-down {
          type yang:date-and-time;
          description
            "The date and time the RPKI-RTR session last
             transitioned in or out of the UP state.";
```

Commenté [MB38]: How the various times are supposed to be used? For example, is there any correlation that need to be followed to detect anomalies?

Commenté [MB39]: There is no such state in the list of state above.

```
                reference
                  "RFC 6810: The Resource Public Key Infrastructure
                              (RPKI) to Router Protocol";
              }
              leaf last-update-sync {
                type yang:date-and-time;
                description
                  "The date and time of the last serial sync with the
                   cache server.";
              }
              leaf last-full-sync {
                type yang:date-and-time;
                description
                  "The date and time of the last reset sync with the
                   cache server.";
              }
              leaf last-serial-query {
                type yang:date-and-time;
                description
                  "The date and time the last serial query was sent to
                   the cache server.";
              }
              leaf last-reset-query {
                type yang:date-and-time;
                description
                  "The date and time the last reset query was sent to
                   the cache server.";
              }
              leaf last-eod-received {
                type yang:date-and-time;
                description
                  "The date and time the last End of Data (EOD) was
received.";
              }
              leaf last-config-change {
                type yang:date-and-time;
                description
                  "The date and time of the last host, port, VRF or
                   local interface change.";
              }
              leaf last-error {
                type yang:date-and-time;
                description
                  "The date and time of the last protocol error sent to
                   or received from the cache server.";
              }
              leaf last-connection-error {
                type yang:date-and-time;
                description
```

Commenté [MB40]: Please add the section where this is discussed

```
                "The date and time of the last connection error to
                 the cache server.";
              }
              leaf last-connection {
                type yang:date-and-time;
                description
                  "The date and time of the last connection to the
                   cache server.";
              }
              leaf error-reason {
                type string;
                description
                  "Reason for error in connection.";
              }
            }
            container protocol-data {
              description
                "Parameters related to the RTR protocol.";
              leaf preferred-protocol-version {
                type uint8;
                description
                  "The preferred version number of the RTR protocol
                   to be used when establishing the session. If not
                   configured, the router will use its default
                   protocol version.";
              }
              leaf refresh-interval {
                type uint32 {
                  range "1..86400";
                }
                default 3600;
                description
                  "Number of seconds a router waits in between
                   sending periodic serial queries to the RPKI
                   cache server. If configured, this value
                   overrides the value announced by the cache
                   server in the End of Data PDU.";
                reference
                  "RFC ZZZZ: The Resource Public Key Infrastructure
                              (RPKI) to Router Protocol, Version 2,
                              Section 6";
              }
              leaf retry-interval {
                type uint32 {
                  range "1..7200";
                }
                default 600;
                description
```

Commenté [MB41]: How this is populated? Is the last observed error for the connection? Else?

Commenté [MB42]: Please add a units statement

Commenté [MB43]: Add units statement

```
                "Number of seconds between poll error and cache
                 server poll. If configured, this value
                 overrides the value announced by the cache
                 server in the End of Data PDU.";
              reference
                "RFC ZZZZ: The Resource Public Key Infrastructure
                           (RPKI) to Router Protocol, Version 2,
                           Section 6";
            }
            leaf expire-interval {
              type uint32 {
                range "600..172800";
              }
              default 7200;
              description
                "Number of seconds to retain data synced from
                 cache server. If configured, this value
                 overrides the value announced by the cache
                 server in the End of Data PDU.";
              reference
                "RFC ZZZZ: The Resource Public Key Infrastructure
                           (RPKI) to Router Protocol, Version 2,
                           Section 6";
            }
            leaf hold-interval {
              type uint32;
              description
                "Maximum number of seconds to wait without receiving
                 any RTR PDUs from the cache server before tearing
                 down the RTR session. Any incoming PDU resets this
                 timer. The hold-interval value must MUST be at least
                 twice the configured refresh-interval. If not
                 configured, implementations should choose a value
                 that is at least twice the configured
                 refresh-interval.";
            }
            leaf protocol-version {
              type uint8;
              config false;
              description
                "The negotiated version number of the RTR protocol
                 currently in use for this session.";
            }
            leaf refresh-time {
              type yang:date-and-time;
              config false;
              description
                "The date and time when the next refresh of RPKI
```

Commenté [MB44]: Add missing units statement

Commenté [MB45]: Add units + reference statements

```
          data from the RPKI cache server will occur.";
      }
      leaf response-time {
        type yang:date-and-time;
        config false;
        description
          "The date and time when a response to a serial or
           reset query to the RPKI cache server is expected.";
      }
      leaf record-lifetime {
        type yang:date-and-time;
        config false;
        description
          "The date and time when the records received from
           the RPKI cache server will expire.";
      }
      leaf hold-time {
        type yang:date-and-time;
        config false;
        description
          "The date and time when the RTR session will be torn
           down if no RTR PDUs are received from the cache
           server.";
      }
      leaf session-id {
        type uint16;
        config false;
        description
          "When a cache server is started, it generates a
           Session ID to identify the instance of the cache
           and to bind it to the sequence of Serial Numbers
           that cache instance will generate.";
        reference
          "RFC 6810: The Resource Public Key Infrastructure
                     (RPKI) to Router Protocol, Section 5.1
           RFC 8210: The Resource Public Key Infrastructure
                     (RPKI) to Router Protocol, Version 1";
      }
      leaf serial-full {
        type uint32;
        config false;
        description
          "A 32-bit strictly increasing unsigned integer which
           wraps from 2^32-1 to 0. It denotes the logical
           version of a cache. It resembles the latest full
           query.";
        reference
          "RFC 6810: The Resource Public Key Infrastructure
```

Commenté [MB46]: Please double check + add a section for 8210

```
                          (RPKI) to Router Protocol, Section X
                    RFC 8210: The Resource Public Key Infrastructure
                          (RPKI) to Router Protocol, Version 1";
              }
              leaf serial-incremental {
                type uint32;
                config false;
                description
                  "A 32-bit strictly increasing unsigned integer which
                   wraps from 2^32-1 to 0. It denotes the logical
                   version of a cache. It resembles the latest
                   incremental query.";
                reference
                  "RFC 6810: The Resource Public Key Infrastructure
                          (RPKI) to Router Protocol, Section X
                   RFC 8210: The Resource Public Key Infrastructure
                          (RPKI) to Router Protocol, Version 1,
Section X";
              }
            }
            container pdu-counters {
              config false;
              description
                "Counters of RTR PDUs exchanged with the RPKI cache
                 server, by PDU type.";
              leaf serial-notify {
                type yang:zero-based-counter64;
                description
                  "Serial Notify (PDU Type 0) count.";
              }
              leaf serial-query {
                type yang:zero-based-counter64;
                description
                  "Serial Query (PDU Type 1) count.";
              }
              leaf reset-query {
                type yang:zero-based-counter64;
                description
                  "Reset Query (PDU Type 2) count.";
              }
              leaf cache-response {
                type yang:zero-based-counter64;
                description
                  "Cache Response (PDU Type 3) count.";
              }
              leaf ipv4-prefix {
                type yang:zero-based-counter64;
                description
                  "IPv4 Prefix (PDU Type 4) count.";
```

Commenté [MB47]: Please add a section. Idem for 8210

a mis en forme : Français (France)

Commenté [MB48]: Please add a section. Idem for 8210

a mis en forme : Français (France)

a mis en forme : Français (France)

Commenté [MB49]: Instead of having a flat structure, what about using a list of errors indexed by types?

The type can be taken from the IANA registry

```
            }
            leaf ipv6-prefix {
              type yang:zero-based-counter64;
              description
                "IPv6 Prefix (PDU Type 6) count.";
            }
            leaf end-of-data {
              type yang:zero-based-counter64;
              description
                "End of Data (PDU Type 7) count.";
            }
            leaf cache-reset {
              type yang:zero-based-counter64;
              description
                "Cache Reset (PDU Type 8) count.";
            }
            leaf router-key {
              type yang:zero-based-counter64;
              description
                "Router Key (PDU Type 9) count.";
            }
            leaf aspa {
              type yang:zero-based-counter64;
              description
                "ASPA (PDU Type 11) count.";
            }
          }
          container error-pdu-counters {
            config false;
            description
              "Counters of Error Report PDUs exchanged with the
               RPKI cache server, by error code.";
            leaf corrupt-data {
              type yang:zero-based-counter64;
              description
                "Corrupt Data (Error Code 0) Error Report PDU
                 count.";
            }
            leaf internal-error {
              type yang:zero-based-counter64;
              description
                "Internal Error (Error Code 1) Error Report PDU
                 count.";
            }
            leaf no-data-available {
              type yang:zero-based-counter64;
              description
                "No Data Available (Error Code 2) Error Report PDU
```

Commenté [MB50]: Instead of having a flat structure, what about using a list of errors indexed by types?

The type can be taken from the IANA registry

```
          count.";
        }
        leaf invalid-request {
          type yang:zero-based-counter64;
          description
            "Invalid Request (Error Code 3) Error Report PDU
             count.";
        }
        leaf unsupported-protocol-version {
          type yang:zero-based-counter64;
          description
            "Unsupported Protocol Version (Error Code 4) Error
             Report PDU count.";
        }
        leaf unsupported-pdu-type {
          type yang:zero-based-counter64;
          description
            "Unsupported PDU Type (Error Code 5) Error Report
             PDU count.";
        }
        leaf withdrawal-unknown-record {
          type yang:zero-based-counter64;
          description
            "Withdrawal of Unknown Record (Error Code 6) Error
             Report PDU count.";
        }
        leaf duplicate-announcement-received {
          type yang:zero-based-counter64;
          description
            "Duplicate Announcement Received (Error Code 7)
             Error Report PDU count.";
        }
        leaf unexpected-protocol-version {
          type yang:zero-based-counter64;
          description
            "Unexpected Protocol Version (Error Code 8) Error
             Report PDU count.";
        }
        leaf aspa-provider-list-error {
          type yang:zero-based-counter64;
          description
            "ASPA Provider List Error (Error Code 9) Error
             Report PDU count.";
        }
        leaf transport-failure {
          type yang:zero-based-counter64;
          description
            "Transport Failure (Error Code 10) Error Report
```

```
                    PDU count.";
                }
                leaf ordering-error {
                  type yang:zero-based-counter64;
                  description
                    "Ordering Error (Error Code 11) Error Report PDU
                     count.";
                }
                leaf cache-restart {
                  type yang:zero-based-counter64;
                  description
                    "Cache Restart (Error Code 12) Error Report PDU
                     count.";
                }
                leaf cache-shutdown {
                  type yang:zero-based-counter64;
                  description
                    "Cache Shutdown (Error Code 13) Error Report PDU
                     count.";
                }
              }
            }
          }
        }
      }
    }
  }
  <CODE ENDS>
```

5. RPKI Table YANG Module

5.1. Tree View

The full tree of the "ietf-rpki-table" YANG module is provided in Appendix B. The following subsections list the subtree structures.

5.1.1. Overall Structure

The overall tree structure of the "ietf-rpki-table" YANG module is shown in Figure 12.

The "ietf-rpki-table" YANG module also augments the core routing YANG module "ietf-routing" specified in [RFC8349]. Specifically, this YANG module augments "/rt:routing".

```
module: ietf-rpki-table
  augment /rt:routing:
    +--ro vrp-tables
    |  +--ro vrp-table* [name]
    |     ...
    +--ro router-key-tables
    |  +--ro router-key-table* [name]
    |     ...
    +--ro aspa-tables
       +--ro aspa-table* [name]
          ...
```

Figure 12: Overall RPKI Table Tree Structure

The "ietf-rpki-table" YANG module has three containers: 'vrp-tables', 'router-key-tables', and 'aspa-tables'.

The 'vrp-tables' container lists the VRP records received from all RPKI cache servers, and each list 'vrp-table' identifies a VRP record table received from a RPKI cache server.

The 'router-key-tables' container lists the Router Key records received from all RPKI cache servers, and each list 'router-key-table' identifies a Router Key record table received from a RPKI cache server.

The 'aspa-tables' container lists the ASPA records received from all RPKI cache servers, and each list 'aspa-table' identifies an ASPA record table received from a RPKI cache server.

5.1.2. VRP Table

The structure of the list 'vrp-table' is shown in Figure 13.

```
module: ietf-rpki-table
  augment /rt:routing:
    +--ro vrp-tables
    |  +--ro vrp-table* [name]
    |     +--ro name                        string
    |     +--ro ccr-roapayloadstate-hash? ccr-hash
    |     +--ro ipv4
    |     |  +--ro vrps
    |     |  |  +--ro vrp* [prefix max-len asn source]
    |     |  |     +--ro prefix       inet:ipv4-prefix
    |     |  |     +--ro max-len      ipv4-prefix-length
    |     |  |     +--ro asn          inet:as-number
    |     |  |     +--ro source       union
    |     |  +--ro total-records?     yang:gauge32
    |     |  +--ro records-added?     yang:counter64
    |     |  +--ro records-deleted?   yang:counter64
    |     +--ro ipv6
    |        +--ro vrps
    |        |  +--ro vrp* [prefix max-len asn source]
    |        |     +--ro prefix       inet:ipv6-prefix
    |        |     +--ro max-len      ipv6-prefix-length
    |        |     +--ro asn          inet:as-number
    |        |     +--ro source       union
    |        +--ro total-records?     yang:gauge32
    |        +--ro records-added?     yang:counter64
    |        +--ro records-deleted?   yang:counter64
    +--ro router-key-tables
    |  +--ro router-key-table* [name]
    |     ...
    +--ro aspa-tables
       +--ro aspa-table* [name]
          ...
```

Figure 13: RPKI VRP Table Subtree Structure

The 'vrp-table' data nodes are described as follows:

'name': Indicates ~~the~~ a name that uniquely identifies ~~of the~~a VRP table among available VRP tables under /rt:routing.

'ccr-roapayloadstate-hash': Indicates the CCR state hash for the ROA payload.

'ipv4': Indicates detailed information and statistics of IPv4 VRP records in the VRP table.

Commenté [MB51]: Why this type is used compared to gauge64?

Commenté [MB52]: Why this type is used compared to gauge64?

Commenté [MB53]: What about defining a feature for aspa. This would allow for modularity

a mis en forme : Anglais (États-Unis)

Commenté [MB54]: Add a reference to draft ccr

This data node includes 'vrps', 'total-records', 'records-added', and 'records-deleted'. The 'vrps' indicates all IPv4 VRP records in the VRP table, and each list 'vrp' indicates an IPv4 VRP record, which includes the IPv4 prefix ('prefix'), the maximum IPv4 prefix length ('max-len'), the origin AS number of the IPv4 prefix ('asn'), and the source of the IPv4 VRP record ('source'). The 'total-records' indicates the total number of IPv4 VRP records. The 'records-added' indicates the number of IPv4 VRP records cumulatively added. The 'records-deleted' indicates the number of IPv4 VRP records cumulatively deleted.

'ipv6': Indicates detailed information and statistics of IPv6 VRP records in the VRP table.

This data node has the same structure as the data node 'ipv4'~~.~~ ~~The 'vrps' indicates all IPv6 VRP records in the VRP table, and each list 'vrp' indicates an IPv6 VRP record, which includes the IPv6 prefix ('prefix'), the maximum IPv6 prefix length ('max-len'), the origin AS number of the IPv6 prefix ('asn'), and the source of the IPv6 VRP record ('source'). The 'total-records' indicates the total number of IPv6 VRP records. The 'records-added' indicates the number of IPv6 VRP records cumulatively added. The 'records-deleted' indicates the number of IPv6 VRP records cumulatively deleted.~~, except that it covers IPv6 addresses.

5.1.3. Router Key Table

The structure of the list 'router-key-table' is shown in Figure 14.

```
module: ietf-rpki-table
  augment /rt:routing:
    +--ro vrp-tables
    |  +--ro vrp-table* [name]
    |     ...
    +--ro router-key-tables
    |  +--ro router-key-table* [name]
    |     +--ro name                      string
    |     +--ro ccr-routerkeystate-hash? ccr-hash
    |     +--ro router-keys
    |        +--ro router-key* [ski asn key server-address]
    |           +--ro ski            subject-key-id
    |           +--ro asn            inet:as-number
    |           +--ro key            binary
    |           +--ro server-address inet:ip-address
    +--ro aspa-tables
       +--ro aspa-table* [name]
          ...
```

Figure 14: RPKI Router Key Table Subtree Structure

The 'router-key-table' data nodes are described as follows:

'name': Indicates the name of the Router Key table.

'ccr-routerkeystate-hash': Indicates the CCR state hash for the Router Key payload.

'router-keys': Indicates detailed information of Router Key records in the Router Key table.

This data node only includes a list 'router-key', which contains the Subject Key Identifier (SKI) of the Router Key ('ski'), the AS number of the router which the Router Key belongs to ('asn'), the subjectPublicKeyInfo value of the Router Key (see [RFC8608]) ('key'), and the IP address of the RPKI cache server ('server-address').

Commenté [MB55]: Can we please reference the exact section?

5.1.4. ASPA Table

The structure of the list 'aspa-table' is shown in Figure 15.

```
module: ietf-rpki-table
  augment /rt:routing:
    +--ro vrp-tables
    |  +--ro vrp-table* [name]
    |     ...
    +--ro router-key-tables
    |  +--ro router-key-table* [name]
    |     ...
    +--ro aspa-tables
       +--ro aspa-table* [name]
          +--ro name                        string
          +--ro ccr-aspapayloadstate-hash? ccr-hash
          +--ro aspas* [customer-asn]
             +--ro customer-asn        inet:as-number
             +--ro server-address?     inet:ip-address
             +--ro provider-asns* [provider-asn]
                +--ro provider-asn     inet:as-number
```

Figure 15: RPKI ASPA Table Subtree Structure

The 'aspa-table' data nodes are described as follows:

'name': Indicates the name of the ASPA table.

'ccr-aspapayloadstate-hash': Indicates the CCR state hash for the ASPA payload.

'aspas': Indicates detailed information of ASPA records in the ASPA table.

This data node includes a leaf 'customer-asn', a leaf 'server-address', and a list 'provider-asns'. The 'customer-asn' indicates the AS number of a customer. The 'server-address' indicates the IP address of the RPKI cache server from which ASPA records are received. The list 'provider-asns' contains a leaf 'provider-asn' which indicates the AS number of a provider of the customer.

5.2. YANG Module

This YANG module has normative references to [RFC6487], [RFC8349], [RFC8608], [RFC9911], [I-D.ietf-sidrops-8210bis], and [I-D.ietf-sidrops-rpki-ccr].

RFC Editor Note: ~~that the~~ "RFC YYYY" in this YANG module must be replaced by the publication version of [I-D.ietf-sidrops-rpki-ccr], and the "RFC ZZZZ" must be replaced by the publication version of [I-D.ietf-sidrops-8210bis]. This note is to be removed after the publication of [I-D.ietf-sidrops-rpki-ccr] and [I-D.ietf-sidrops-8210bis].

Commenté [MB56]: You may group all notes to the RFC Editor in one place.

```
<CODE BEGINS> file "ietf-rpki-table@2026-10-15.yang"
module ietf-rpki-table {
  yang-version 1.1;
  namespace "urn:ietf:params:xml:ns:yang:ietf-rpki-table";
  prefix rpki-table;

  import ietf-yang-types {
    prefix yang;
    reference
      "RFC 9911: Common YANG Data Types, Section 3";
  }
  import ietf-inet-types {
    prefix inet;
    reference
      "RFC 9911: Common YANG Data Types, Section 4";
  }
  import ietf-routing {
    prefix rt;
    reference
      "RFC 8349: A YANG Data Model for Routing Management
                 (NMDA Version)";
```

```
    }

    organization
      "IETF Secure Inter-Domain Routing Operations (SIDROPS)
       Working Group";
    contact
      "WG Web:   <https://datatracker.ietf.org/wg/sidrops/>
       WG List:  SIDROPS <mailto:sidrops@ietf.org>

       EditorAuthor:   Yisong Liu
                 <mailto:liuyisong@chinamobile.com>
       AuthorEditor:   Changwang Lin
                 <mailto:linchangwang.04414@h3c.com>
       AuthorEditor:   Haibo Wang
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                 <mailto:jishnu.roy@hpe.com>
       AuthorEditor:   Jeffrey Haas
                 <mailto:jeffrey.haas@hpe.com>
       AuthorEditor:   Hongwei Liu
                 <mailto:liu.hongwei3@zte.com.cn>
       AuthorEditor:   Di Ma
                 <mailto:madi@zdns.cn>";
    description
      "This module describes a YANG model module for the Resource Public
       Key Infrastructure (RPKI) to Router (RTR) protocol data
       management.

       The key words 'MUST', 'MUST NOT', 'REQUIRED', 'SHALL', 'SHALL
       NOT', 'SHOULD', 'SHOULD NOT', 'RECOMMENDED', 'NOT RECOMMENDED',
       'MAY', and 'OPTIONAL' in this document are to be interpreted as
       described in BCP 14 (RFC 2119) (RFC 8174) when, and only when,
       they appear in all capitals, as shown here.

       Copyright (c) 2026 IETF Trust and the persons identified as
       authors of the code. All rights reserved.

       Redistribution and use in source and binary forms, with or
       without modification, is permitted pursuant to, and subject to
       the license terms contained in, the Revised BSD License set
       forth in Section 4.c of the IETF Trust's Legal Provisions
       Relating to IETF Documents
       (https://trustee.ietf.org/license-info).

       All revisions of IETF and IANA published modules can be found
       at the YANG Parameters registry group
       (https://www.iana.org/assignments/yang-parameters).
```

Commenté [MB57]: Per the template in 9907

a mis en forme : Anglais (États-Unis)

Commenté [MB58]: This description is almost identical to the first module. Consider updating to reflect the content.

```
        This version of this YANG module is part of RFC XXXX; see the
        RFC itself for full legal notices.";

    revision 2026-10-15 {
      description
        "Initial Version";
      reference
        "RFC XXXX: A YANG Data Model for RPKI to Router Protocol";
    }

    typedef ipv4-prefix-length {
      type uint8 {
        range "0..32";
      }
      description
        "IPv4 Prefix Length.";
    }

    typedef ipv6-prefix-length {
      type uint8 {
        range "0..128";
      }
      description
        "IPv6 Prefix Length.";
    }

    typedef subject-key-id {
      type binary {
        length "20";
      }
      description
        "Subject Key Identifier.";
    }

    typedef ccr-hash {
      type binary {
        length "32";
      }
      description
        "Canonical Cache Representation (CCR) state hash.

         This represents a 256-bit SHA-256 hash of the canonical
         representation of RPKI data as defined in RFC YYYY.

         The hash provides a compact, verifiable representation
         of the complete state of RPKI validation data, enabling
         efficient synchronization and consistency verification
         between caches.";
```

Commenté [MB59]: To reflect the exact title

```
      reference
        "RFC YYYY: A Profile for Resource Public Key Infrastructure
                   (RPKI) Canonical Cache Representation (CCR)";
    }

    grouping aspa-overall-records {
      description
        "Autonomous System Provider Authorization (ASPA) records
         received from all RPKI cache servers.";
      reference
        "RFC ZZZZ: The Resource Public Key Infrastructure (RPKI) to
                   Router Protocol, Version 2";
      list aspas {
        key "customer-asn";
        description
          "An entry for ASPA.";
        leaf customer-asn {
          type inet:as-number;
          description
            "The AS number of a customer.";
        }
        leaf server-address {
          type inet:ip-address;
          description
            "IP address of the RPKI cache server.";
        }
        list provider-asns {
          key "provider-asn";
          description
            "Providers of the customer.";
          leaf provider-asn {
            type inet:as-number;
            description
              "The AS number of a provider.";
          }
        }
      }
    }

    augment "/rt:routing" {
      description
        "RPKI tables augmentation of ietf-routing module.";
      container vrp-tables {
        config false;
        description
          "List of tables containing Validated ROA Payloads
           received from all RPKI cache servers.";
        list vrp-table {
```

```
          key "name";
          description
            "Table of Validated ROA Payloads received from
             a RPKI cache server.";
          leaf name {
            type string;
            description
              "Name of the Validated ROA Payload table.";
          }
          leaf ccr-roapayloadstate-hash {
            type ccr-hash;
            description
              "CCR state hash for the ROA payload.

               This hash represents the canonical state of this
               specific ROA entry as defined in the CCR specification.
               It is optional and can be used to verify the integrity
               and consistency of ROA data originating from the RPKI
               cache.";
            reference
              "RFC YYYY: A Profile for Resource Public Key
                         Infrastructure (RPKI) Canonical Cache
                         Representation (CCR)";
          }
          container ipv4 {
            config false;
            description
              "Container for IPv4 Validated ROA Payloads table.";
            container vrps {
              config false;
              description
                "IPv4 Validated ROA Payloads received from the RPKI
                 cache server.";
              list vrp {
                key "prefix max-len asn source";
                description
                  "An entry of IPv4 Validated ROA Payload.";
                leaf prefix {
                  type inet:ipv4-prefix;
                  description
                    "The IPv4 prefix of the IPv4 Validated ROA
                     Payload.";
                }
                leaf max-len {
                  type ipv4-prefix-length;
                  description
                    "Denotes the longest IPv4 prefix allowed. This
                     MUST NOT be less than the IPv4 prefix length.";
```

Commenté [MB60]: Which prefix? The one in «prefix»?

Please make that clear

```
                }
                leaf asn {
                  type inet:as-number;
                  description
                    "The origin AS number of the IPv4 Validated ROA
                     Payload.";
                }
                leaf source {
                  type union {
                    type inet:ip-address;
                    type string;
                  }
                  description
                    "String representing the source of the record
                     in this record-set.";
                }
              }
            }
            leaf total-records {
              type yang:gauge32;
              description
                "Total number of IPv4 Validated ROA Payload records.";
            }
            leaf records-added {
              type yang:counter64;
              description
                "Number of IPv4 Validated ROA Payload records
                 cumulatively added.";
            }
            leaf records-deleted {
              type yang:counter64;
              description
                "Number of IPv4 Validated ROA Payload records
                 cumulatively deleted.";
            }
          }
          container ipv6 {
            config false;
            description
              "Container for IPv6 Validated ROA Payloads table.";
            container vrps {
              config false;
              description
                "IPv6 Validated ROA Payloads received from the RPKI
                 cache server.";
              list vrp {
                key "prefix max-len asn source";
                description
```

Commenté [MB61]: Why isn't this IPv4?

Commenté [MB62]: This can also be an IP address. Please adjust the description to match the type

Commenté [MB63]: May save some space if this was defined as a grouping as the same structure is used for both v4 and v6. This is very minor though

```
                "An entry for IPv6 Validated ROA Payload.";
              leaf prefix {
                type inet:ipv6-prefix;
                description
                  "The IPv6 prefix of the IPv6 Validated ROA
                   Payload.";
              }
              leaf max-len {
                type ipv6-prefix-length;
                description
                  "Denotes the longest IPv6 prefix allowed. This
                   MUST NOT be less than the prefix length.";
              }
              leaf asn {
                type inet:as-number;
                description
                  "The origin AS number of the IPv6 Validated ROA
                   Payload.";
              }
              leaf source {
                type union {
                  type inet:ip-address;
                  type string;
                }
                description
                  "Representing the source of the record in this
                   record-set. Either a server IP or a source file
                   of static records.";
              }
            }
          }
          leaf total-records {
            type yang:gauge32;
            description
              "Total number of IPv6 Validated ROA Payload records.";
          }
          leaf records-added {
            type yang:counter64;
            description
              "Number of IPv6 Validated ROA Payload records
               cumulatively added.";
          }
          leaf records-deleted {
            type yang:counter64;
            description
              "Number of IPv6 Validated ROA Payload records
               cumulatively deleted.";
          }
```

Commenté [MB64]: Same comment as for the IPv4 data node above

Commenté [MB65]: Same comment as for IPv4

Commenté [MB66]: This is better compared to the description for IPv4 similar node

Commenté [MB67]: What about tracking discontinuity time?

The same comment applies for the other tables.

```
          }
        }
      }
      container router-key-tables {
        config false;
        description
          "List of Router Key table received from all RPKI cache
           servers.";
        list router-key-table {
          key "name";
          description
            "Table of Router Keys received from a RPKI cache
             server.";
          leaf name {
            type string;
            description
              "Name of the Router Key table.";
          }
          leaf ccr-routerkeystate-hash {
            type ccr-hash;
            description
              "CCR state hash for the Router Key payload.

               This hash represents the canonical state of this
               specific Router Key entry as defined in the CCR
               specification. It is optional and can effectively
               verify the consistency of BGPsec Router Key data
               derived from the RPKI cache.";
            reference
              "RFC YYYY: A Profile for Resource Public Key
                         Infrastructure (RPKI) Canonical Cache
                         Representation (CCR)";
          }
          container router-keys {
            config false;
            description
              "Router Keys received from the RPKI cache server.";
            list router-key {
              key "ski asn key server-address";
              description
                "An entry for Router Key.";
              leaf ski {
                type subject-key-id;
                description
                  "A Router Key's Subject Key Identifier (SKI).";
                reference
                  "RFC 6487: A Profile for X.509 PKIX Resource
                             Certificates";
```

```
            }
            leaf asn {
              type inet:as-number;
              description
                "The AS number of the router which the key
                 belongs to.";
            }
            leaf key {
              type binary;
              description
                "A Router Key's subjectPublicKeyInfo value, as
                 described in Section 3.1.1 of RFC 8608, encoded
                 using ASN.1 Distinguished Encoding Rules (DER).";
              reference
                "RFC 8608: BGPsec Algorithms, Key Formats, and
                           Signature Formats";
            }
            leaf server-address {
              type inet:ip-address;
              description
                "IP address of the RPKI cache server.";
            }
          }
        }
      }
    }
    container aspa-tables {
      config false;
      description
        "List of tables of ASPAs received from all RPKI cache
         servers.";
      list aspa-table {
        key "name";
        description
          "Table of ASPAs received from a RPKI cache server.";
        leaf name {
          type string;
          description
            "Name of the ASPA table.";
        }
        leaf ccr-aspapayloadstate-hash {
          type ccr-hash;
          description
            "CCR state hash for the ASPA payload.

             This hash represents the canonical state of this
             specific ASPA entry as defined in the CCR
             specification. It is optional and facilitates
```

```
                consistency checking for AS path validation data
                originating from the RPKI cache.";
              reference
                "RFC YYYY: A Profile for Resource Public Key
                           Infrastructure (RPKI) Canonical Cache
                           Representation (CCR)";
            }
            uses aspa-overall-records;
          }
        }
      }
    }
  }
  <CODE ENDS>
```

6. Implementation Status

Note to the RFC Editor - remove this section before publication, as well as remove the reference to [RFC7942].

This section records the status of known implementations of the protocol defined by this specification at the time of posting of this Internet-Draft, and is based on a proposal described in [RFC7942]. The description of implementations in this section is intended to assist the IETF in its decision processes in progressing drafts to RFCs. Please note that the listing of any individual implementation here does not imply endorsement by the IETF. Furthermore, no effort has been spent to verify the information presented here that was supplied by IETF contributors. This is not intended as, and must not be construed to be, a catalog of available implementations or their features. Readers are advised to note that other implementations may exist.

According to [RFC7942], "this will allow reviewers and working groups to assign due consideration to documents that have the benefit of running code, which may serve as evidence of valuable experimentation and feedback that have made the implemented protocols more mature. It is up to the individual working groups to use this information as they see fit".

6.1. Juniper Networks (HPE)

* Organization: Juniper Networks (HPE).

* Implementation: The following leaves/parameters in description are implemented.

* Description: YANG model leaves that are supported:

- ii~~i~~etf-rpki-rtr.yang

  ~~Server~~server-address
  ~~Session~~session-state
  ~~Total~~total-vrp-records
  ii~~i~~pv4-vrp-records
  ii~~i~~pv6-vrp-records
  r~~R~~efresh-time
  ~~Hold~~hold-time
  ~~Serial~~serial-incremental
  s~~S~~erial-full
  ll~~L~~ast-session-up-down
  ll~~L~~ast-update-sync
  ll~~L~~ast-eod-received
  ll~~L~~ast-serial-query
  ll~~L~~ast-reset-query
  f~~F~~laps

- ~~Ietf~~ietf-rpki-table.yang

  ~~Ipv4~~ ipv4 | ipv6:
    ~~Total~~total-records
    ~~Records~~records-added
    ~~Records~~records-deleted~~.~~

* Maturity Level: Ready-for-deployment

* Coverage:

* Version: Draft-03

* Licensing: N/A

* Implementation experience: Nothing specific.

* Contact: jishnu.roy@hpe.com

* Last updated: March 30, 2026

6.2. New H3C Technologies

* Organization: New H3C Technologies.

* Implementation: The following leaves/parameters in description are implemented.

* Description: "ietf-rpki-rtr" and "ietf-rpki-table" YANG modules have been implemented in New H3C Products.

* Maturity Level: Ready-for-deployment

* Coverage: All data nodes of "ietf-rpki-rtr" and "ietf-rpki-table" YANG modules.

* Version: Draft-03

* Licensing: N/A

* Implementation experience: Nothing specific.

* Contact: li_meng_limeng@h3c.com

* Last updated: March 30, 2026

7. Security Considerations

This section is modeled after the template described in Section 3.7.1 of [RFC9907].

The "ietf-rpki-rtr" YANG module and "ietf-rpki-table" YANG module define data models that are designed to be accessed via YANG-based management protocols, such as Network Configuration Protocol (NETCONF) [RFC6241] and RESTCONF [RFC8040]. These YANG-based management protocols (1) have to use a secure transport layer (e.g., Secure Shell (SSH) [RFC4252], TLS [RFC9846], and QUIC [RFC9000]) and (2) have to use mutual authentication.

The Network Configuration Access Control Model (NACM) [RFC8341] provides the means to restrict access for particular NETCONF or RESTCONF users to a preconfigured subset of all available NETCONF or RESTCONF protocol operations and content.

There are a number of data nodes defined in these YANG modules that are writable/creatable/deletable (i.e., config true, which is the default). All writable data nodes are likely to be sensitive or vulnerable in some network environments. Write operations (e.g., edit-config) and delete operations to these data nodes without proper protection or authentication can have a negative effect on network operations. The following subtrees and data nodes have particular sensitivities/vulnerabilities:

Commenté [MB68]: Please update to indicate for each of these their particular vulnerabilities.

For example, a very low or very large limit may lead to DoS, etc.

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:server-address

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:server-port

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:local-address

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:local-port

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:enabled

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:preference

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:description

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:enable-authentication

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:authentication

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:vrp-limit

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:aspa-limit

Some of the readable data nodes in these YANG modules may be considered sensitive or vulnerable in some network environments. It is thus important to control read access (e.g., via get, get-config, or notification) to these data nodes. Specifically, the following subtrees and data nodes have particular sensitivities/ vulnerabilities:

Commenté [MB69]: Why is this particulalry vulnerable?

Commenté [MB70]: Idem as above.

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:session-state

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:statistics

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:connection-data

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:protocol-data

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:pdu-counters

rpki-rtr:rpki-rtr/rpki-rtr:sessions/rpki-rtr:session/rpki-rtr:error-pdu-counters

rt:routing/rpki-table:vrp-tables/rpki-table:vrp-table

rt:routing/rpki-table:router-key-tables/rpki-table:router-key-table

rt:routing/rpki-table:aspa-tables/rpki-table:aspa-table

There are no particularly sensitive RPC or action operations.

8. IANA Considerations

~~8.1. RPKI to Router YANG Module Registry~~

IANA is requested to register the following URIs in the "ns" registry within the "IETF XML Registry" group [RFC3688]~~({RFC3688})~~:

URI: urn:ietf:params:xml:ns:yang:ietf-rpki-rtr
Registrant Contact: The IESG.
XML: N/A; the requested URI is an XML namespace

URI: urn:ietf:params:xml:ns:yang:ietf-rpki-table
Registrant Contact: The IESG.
XML: N/A; the requested URI is an XML namespace

IANA is requested to register the following YANG modules in the "YANG Module Names" registry ~~(~~[RFC6020]~~)~~ within the "YANG Parameters" registry group.

Name: ietf-rpki-rtr
Maintained by IANA? N
Namespace: urn:ietf:params:xml:ns:yang:ietf-rpki-rtr
Prefix: rpki-rtr
Reference: RFC XXXX

Name: ietf-rpki-table

```
      Maintained by IANA? N
      Namespace: urn:ietf:params:xml:ns:yang:ietf-rpki-table
      Prefix: rpki-table
      Reference: RFC XXXX
```

~~8.2. RPKI Table YANG Module Registry~~

~~IANA is requested to register the following URI in the "ns" registry within the "IETF XML Registry" group ([RFC3688]):~~

~~URI: urn:ietf:params:xml:ns:yang:ietf-rpki-table~~
~~Registrant Contact: The IESG.~~
~~XML: N/A; the requested URI is an XML namespace~~

~~IANA is requested to register the following YANG module in the "YANG Module Names" registry ([RFC6020]) within the "YANG Parameters" registry group.~~

~~Name: ietf-rpki-table~~
~~Maintained by IANA? N~~
~~Namespace: urn:ietf:params:xml:ns:yang:ietf-rpki-table~~
~~Prefix: rpki-table~~
~~Reference: RFC XXXX~~

9. Acknowledgments

The authors would like to thank Job Snijders, Santosh Kolenchery, Ebben Xavier Aries, Tapasee Ratna Goutam, and Haiyang Zhang for their review and discussion of this document.

Thanks to Per Andersson for the Yangdoctors Early review.

10. References

10.1. Normative References

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Bush, R., Austein, R., and T. Harrison, "The Resource Public Key Infrastructure (RPKI) to Router Protocol, Version 2", Work in Progress, Internet-Draft, draft-ietf-sidrops-8210bis-27, 13 August 2026, <https://datatracker.ietf.org/doc/html/draft-ietf-sidrops-8210bis-27>.

[I-D.ietf-sidrops-rpki-ccr]
Snijders, J., Bakker, B., Bruijnzeels, T., and T. Buehler, "A Profile for Resource Public Key Infrastructure (RPKI) Canonical Cache Representation (CCR)", Work in Progress, Internet-Draft, draft-ietf-sidrops-rpki-ccr-11, 1 July 2026, <https://datatracker.ietf.org/doc/html/draft-ietf-sidrops-rpki-ccr-11>.

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Appendix A. Full Tree of RPKI to Router YANG Module

```
module: ietf-rpki-rtr
  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol:
    +--rw rpki-rtr
       +--rw sessions
          +--rw session* [server-address]
             +--rw server-address           inet:ip-address
             +--rw server-port?             inet:port-number
             +--rw local-address?           union
             +--rw local-port?              inet:port-number
             +--rw enabled?                 boolean
             +--rw preference?              uint32
             +--rw description?             string
             +--rw reconnect-interval?      uint32
             +--ro session-state?           enumeration
             +--rw enable-authentication?   boolean
             +--rw authentication
             |  +--rw (option)?
             |     +--:(md5)
             |     |  +--rw md5-password?   ianach:crypt-hash
             |     +--:(ssh)
             |     |  +--rw client-identity
             |     |  |  +--rw username?  string
             |     |  |  +--rw public-key!
```

```
|     |  |                   {client-ident-publickey}?
|     |  |  +--rw password!
|     |  |                   {client-ident-password}?
|     |  |  +--rw hostbased!
|     |  |                   {client-ident-hostbased}?
|     |  |  +--rw none?      empty {client-ident-none}?
|     |  |  +--rw certificate!
|     |  |                   {sshcmn:ssh-x509-certs}?
|     |  +--rw server-authentication
|     |  |  +--rw ssh-host-keys!
|     |  |  +--rw ca-certs!  {sshcmn:ssh-x509-certs}?
|     |  |  +--rw ee-certs!  {sshcmn:ssh-x509-certs}?
|     |  +--rw transport-params
|     |  |                {sshcmn:transport-params}?
|     |  +--rw keepalives!   {ssh-client-keepalives}?
|     |     +--rw max-wait?       uint16
|     |     +--rw max-attempts?   uint8
|     +--:(tcp-ao-keychain)
|        +--rw keychain-name?  key-chain:key-chain-ref
+--rw vrp-limit
|  +--rw max-number?              uint64
|  +--rw threshold-percentage?    rt-types:percentage
|  +--rw over-threshold-action?   enumeration
+--rw aspa-limit
|  +--rw max-number?              uint64
|  +--rw threshold-percentage?    rt-types:percentage
|  +--rw over-threshold-action?   enumeration
+--ro statistics
|  +--ro total-vrp-records?  yang:zero-based-counter64
|  +--ro ipv4-vrp-records?   yang:zero-based-counter64
|  +--ro ipv6-vrp-records?   yang:zero-based-counter64
|  +--ro router-key-records? yang:zero-based-counter64
|  +--ro aspa-records?       yang:zero-based-counter64
|  +--ro in-total-messages?  yang:zero-based-counter64
|  +--ro out-total-messages? yang:zero-based-counter64
+--ro connection-data
|  +--ro flaps?                       uint32
|  +--ro last-session-up-down?        yang:date-and-time
|  +--ro last-update-sync?            yang:date-and-time
|  +--ro last-full-sync?              yang:date-and-time
|  +--ro last-serial-query?           yang:date-and-time
|  +--ro last-reset-query?            yang:date-and-time
|  +--ro last-eod-received?           yang:date-and-time
|  +--ro last-config-change?          yang:date-and-time
|  +--ro last-error?                  yang:date-and-time
|  +--ro last-connection-error?       yang:date-and-time
|  +--ro last-connection?             yang:date-and-time
|  +--ro error-reason?                string
```

```
               +--rw protocol-data
               |  +--rw preferred-protocol-version?  uint8
               |  +--rw refresh-interval?            uint32
               |  +--rw retry-interval?              uint32
               |  +--rw expire-interval?             uint32
               |  +--rw hold-interval?               uint32
               |  +--ro protocol-version?            uint8
               |  +--ro refresh-time?                yang:date-and-time
               |  +--ro response-time?               yang:date-and-time
               |  +--ro record-lifetime?             yang:date-and-time
               |  +--ro hold-time?                   yang:date-and-time
               |  +--ro session-id?                  uint16
               |  +--ro serial-full?                 uint32
               |  +--ro serial-incremental?          uint32
               +--ro pdu-counters
               |  +--ro serial-notify?    yang:zero-based-counter64
               |  +--ro serial-query?     yang:zero-based-counter64
               |  +--ro reset-query?      yang:zero-based-counter64
               |  +--ro cache-response?   yang:zero-based-counter64
               |  +--ro ipv4-prefix?      yang:zero-based-counter64
               |  +--ro ipv6-prefix?      yang:zero-based-counter64
               |  +--ro end-of-data?      yang:zero-based-counter64
               |  +--ro cache-reset?      yang:zero-based-counter64
               |  +--ro router-key?       yang:zero-based-counter64
               |  +--ro aspa?             yang:zero-based-counter64
               +--ro error-pdu-counters
                  +--ro corrupt-data?      yang:zero-based-counter64
                  +--ro internal-error?    yang:zero-based-counter64
                  +--ro no-data-available? yang:zero-based-counter64
                  +--ro invalid-request?   yang:zero-based-counter64
                  +--ro unsupported-protocol-version?
                  |                        yang:zero-based-counter64
                  +--ro unsupported-pdu-type?
                  |                        yang:zero-based-counter64
                  +--ro withdrawal-unknown-record?
                  |                        yang:zero-based-counter64
                  +--ro duplicate-announcement-received?
                  |                        yang:zero-based-counter64
                  +--ro unexpected-protocol-version?
                  |                        yang:zero-based-counter64
                  +--ro aspa-provider-list-error?
                  |                        yang:zero-based-counter64
                  +--ro transport-failure?  yang:zero-based-counter64
                  +--ro ordering-error?      yang:zero-based-counter64
                  +--ro cache-restart?       yang:zero-based-counter64
                  +--ro cache-shutdown?      yang:zero-based-counter64
```

Appendix B. Full Tree of RPKI Table YANG Module

```
   module: ietf-rpki-table
     augment /rt:routing:
       +--ro vrp-tables
       |  +--ro vrp-table* [name]
       |     +--ro name                        string
       |     +--ro ccr-roapayloadstate-hash? ccr-hash
       |     +--ro ipv4
       |     |  +--ro vrps
       |     |  |  +--ro vrp* [prefix max-len asn source]
       |     |  |     +--ro prefix       inet:ipv4-prefix
       |     |  |     +--ro max-len      ipv4-prefix-length
       |     |  |     +--ro asn          inet:as-number
       |     |  |     +--ro source       union
       |     |  +--ro total-records?     yang:gauge32
       |     |  +--ro records-added?     yang:counter64
       |     |  +--ro records-deleted?   yang:counter64
       |     +--ro ipv6
       |        +--ro vrps
       |        |  +--ro vrp* [prefix max-len asn source]
       |        |     +--ro prefix       inet:ipv6-prefix
       |        |     +--ro max-len      ipv6-prefix-length
       |        |     +--ro asn          inet:as-number
       |        |     +--ro source       union
       |        +--ro total-records?     yang:gauge32
       |        +--ro records-added?     yang:counter64
       |        +--ro records-deleted?   yang:counter64
       +--ro router-key-tables
       |  +--ro router-key-table* [name]
       |     +--ro name                      string
       |     +--ro ccr-routerkeystate-hash? ccr-hash
       |     +--ro router-keys
       |        +--ro router-key* [ski asn key server-address]
       |           +--ro ski             subject-key-id
       |           +--ro asn             inet:as-number
       |           +--ro key             binary
       |           +--ro server-address inet:ip-address
       +--ro aspa-tables
          +--ro aspa-table* [name]
             +--ro name                        string
             +--ro ccr-aspapayloadstate-hash? ccr-hash
             +--ro aspas* [customer-asn]
                +--ro customer-asn        inet:as-number
                +--ro server-address?     inet:ip-address
                +--ro provider-asns* [provider-asn]
                   +--ro provider-asn     inet:as-number
```

XX

Commenté [MB72]: Consider adding some examples per the guidance in https://www.rfc-editor.org/info/rfc9907/#section-3.12

No need to be exhaustive.

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